Appendix T

Project Name	AIBI CV for Manufacturing	
Test Suit	System/Manual Functional	
Test Case ID	TC-SFR1-MAN-01	
What to Test	Live camera detects all visible QR codes.	
Test Data Input	Printed QR codes placed in camera FOV.	
Expected Result	All visible codes appear in UI/output.	
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	SFR1, SFR15, SFR16
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing	
Test Suit	System/Manual Functional	
Test Case ID	TC-SFR2-MAN-01	
What to Test	Operator configures field mapping correctly.	
Test Data Input	Mapping config UI plus known printed QRs.	
Expected Result	Output places values into correct fields.	
Traceability	Relevant User Req.(s)	UFR2
	Relevant System Req.(s)	SFR2
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing
Test Suit	System/Manual Functional
Test Case ID	TC-SFR4-MAN-01
What to Test	Real scan produces correct JSON record.
Test Data Input	Live scan event with known QRs
Expected	One JSON record created with all data and metadata.

Result		
Traceability	Relevant User Req.(s)	UFR1, UFR4
	Relevant System Req.(s)	SFR4, SFR11
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	System/Manual Functional	
Test Case ID	TC-SFR5-MAN-01	
What to Test	Keyboard-emulation outputs correct text to target app.	
Test Data Input	Live scan and notepad field in focus.	
Expected Result	Exactly correct keystroke output appears.	
Traceability	Relevant User Req.(s)	UFR9
	Relevant System Req.(s)	SFR5, SFR18
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	System/Manual Functional	
Test Case ID	TC-SFR11-MAN-01	
What to Test	Persistence and replay after downstream outage.	
Test Data Input	Perform scans while downstream offline, then reconnect.	
Expected Result	All offline events replay once with no loss or duplicates.	
Traceability	Relevant User Req.(s)	UFR7
	Relevant System Req.(s)	SFR11
	Relevant Use Case.(s)	UC-002, UC-004

Project Name	**AIBI CV for Manufacturing**
Test Suit	System/Manual Functional
Test Case ID	TC-SFR12-MAN-01

What to Test	Documentation matches schema and outputs.	
Test Data Input	JSON schema doc and sample outputs	
Expected Result	Documentation consistent with schema/output.	
Traceability	Relevant User Req.(s)	UFR4
	Relevant System Req.(s)	SFR12
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	System/Manual Functional	
Test Case ID	TC-SFR15-MAN-01	
What to Test	Live multi-code decoding in typical lighting.	
Test Data Input	Printed QR sheet with known values	
Expected Result	Al codes decode correctly.	
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	SFR15
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	System/Manual Functional	
Test Case ID	TC-SFR16-MAN-01	
What to Test	Output includes all visible QRs	
Test Data Input	Live multi-code scan.	
Expected Result	Output matches count and content of visible QRs	
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	SFR16
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing	
Test Suit	System/Manual Functional	
Test Case ID	TC-SFR17-MAN-01	
What to Test	Developer runs simulation scenario via UI.	
Test Data Input	Simulation UI and recorded footage.	
Expected Result	Results mimic live scan behavior; logs produced.	
Traceability	Relevant User Req.(s)	UFR11
	Relevant System Req.(s)	SFR17
	Relevant Use Case.(s)	UC-005

Project Name	AIBI CV for Manufacturing	
Test Suit	System/Manual Functional	
Test Case ID	TC-SFR18-MAN-01	
What to Test	Simulation produces correct keyboard-emulation output.	
Test Data Input	Simulation scenario and sandbox text field.	
Expected Result	Typed text matches expected sequence.	
Traceability	Relevant User Req.(s)	UFR11
	Relevant System Req.(s)	SFR18
	Relevant Use Case.(s)	UC-005

Project Name	AIBI CV for Manufacturing	
Test Suit	Usability and Operator Workflow	
Test Case ID	TC-PUSR1-MAN-01	
What to Test	New operator completes basic workflow in less than 30 minutes.	
Test Data Input	New user, system, UI, stopwatch	
Expected Result	Operator completes tasks without help in less than or equal to 30 minutes.	
Traceability	Relevant User Req.(s)	UFR8, PUUR1

	Relevant System Req.(s)	PUSR1
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing	
Test Suit	Usability and Operator Workflow	
Test Case ID	TC-PUSR2-MAN-01	
What to Test	UI tooltips/guidance allow correct interpretation.	
Test Data Input	UI and test user	
Expected Result	User identifies major controls based on guidance.	
Traceability	Relevant User Req.(s)	PUUR1
	Relevant System Req.(s)	PUSR2
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing	
Test Suit	Usability and Operator Workflow	
Test Case ID	TC-PUSR3-MAN-01	
What to Test	Workflow requires fewer steps than legacy.	
Test Data Input	Legacy workflow vs. new workflow comparison.	
Expected Result	New system uses fewer steps or requires less manual input	
Traceability	Relevant User Req.(s)	UFR8
	Relevant System Req.(s)	PUSR3
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing
Test Suit	Usability and Operator Workflow
Test Case ID	TC-PUSR4-MAN-01
What to Test	Touch-free operation of normal workflow.
Test Data Input	Operator performing job with system

Expected Result	Operator rarely or never touches UI during typical cycle.	
Traceability	Relevant User Req.(s)	PUUR2, UFR8
	Relevant System Req.(s)	PUSR4
	Relevant Use Case.(s)	UC-002

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Usability and Operator Workflow	
Test Case ID	TC-PUSR5-MAN-01	
What to Test	Automation reduces repetitive manual input.	
Test Data Input	Several consecutive jobs.	
Expected Result	Common fields auto-filled correctly.	
Traceability	Relevant User Req.(s)	PUUR2, UFR8
	Relevant System Req.(s)	PUSR5
	Relevant Use Case.(s)	UC-001, UC-002

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Usability and Operator Workflow	
Test Case ID	TC-PUSR10-MAN-01	
What to Test	Users interpret icons/colors without text.	
Test Data Input	Screenshots presented to users.	
Expected Result	Users correctly identify states (OK/warn/error)	
Traceability	Relevant User Req.(s)	PUUR4, PUUR1, UFR8
	Relevant System Req.(s)	PUSR10
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**
Test Suit	System-Level Non-Functional

Test Case ID	TC-PPSR2-MAN-01	
What to Test	Real-time feedback is “immediate”.	
Test Data Input	Live tasks monitored by operator.	
Expected Result	Operator does not perceive delay.	
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	PPSR2
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	System-Level Non-Functional	
Test Case ID	TC-PPSR5-MAN-01	
What to Test	System works reliably with 720p camera.	
Test Data Input	720p hardware test.	
Expected Result	No noticeable accuracy degradation.	
Traceability	Relevant User Req.(s)	UFR3
	Relevant System Req.(s)	PPSR5
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	System-Level Non-Functional	
Test Case ID	TC-PDSR1-MAN-01	
What to Test	No outbound network traffic during operation.	
Test Data Input	Network monitor during typical shift.	
Expected Result	No data sent externally.	
Traceability	Relevant User Req.(s)	UFR10, UFR7
	Relevant System Req.(s)	PDSR1
	Relevant Use Case.(s)	UC-001, UC-002

Project Name	AIBI CV for Manufacturing	
Test Suit	System-Level Non-Functional	
Test Case ID	TC-PDSR2-MAN-01	
What to Test	System operates with no internet.	
Test Data Input	Disconnect internet; run system	
Expected Result	Full operation continues normally.	
Traceability	Relevant User Req.(s)	UFR10, UFR11
	Relevant System Req.(s)	PDSR2
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing	
Test Suit	System-Level Non-Functional	
Test Case ID	TC-PDSR4-MAN-01	
What to Test	Stable operation under factory conditions.	
Test Data Input	Pilot deployment in factory.	
Expected Result	No crashes; acceptable performance.	
Traceability	Relevant User Req.(s)	UFR11
	Relevant System Req.(s)	PDSR4
	Relevant Use Case.(s)	UC-002

Project Name	AIBI CV for Manufacturing	
Test Suit	System-Level Non-Functional	
Test Case ID	TC-ODSR2-MAN-01	
What to Test	All OSS components documented.	
Test Data Input	Documentation review.	
Expected Result	Each installed library listed with name/version/license.	
Traceability	Relevant User Req.(s)	ODUR1

	Relevant System Req.(s)	ODSR2
	Relevant Use Case.(s)	UC-005

Project Name	AIBI CV for Manufacturing	
Test Suit	System-Level Non-Functional	
Test Case ID	TC-OOSR4-MAN-01	
What to Test	Legacy data imports correctly in real scenario.	
Test Data Input	Actual legacy system export file.	
Expected Result	Records appear correctly with no manual cleanup.	
Traceability	Relevant User Req.(s)	OOUR3
	Relevant System Req.(s)	OOSR4
	Relevant Use Case.(s)	UC-004

Project Name	AIBI CV for Manufacturing	
Test Suit	System-Level Non-Functional	
Test Case ID	TC-EISR4-MAN-01	
What to Test	JSON schema version-controlled and documented.	
Test Data Input	Repo review and documentation.	
Expected Result	Schema file tracked; docs match version.	
Traceability	Relevant User Req.(s)	EIUR2
	Relevant System Req.(s)	EISR4
	Relevant Use Case.(s)	UC-005

Project Name	AIBI CV for Manufacturing
Test Suit	System-Level Non-Functional
Test Case ID	TC-EISR5-MAN-01
What to Test	Export to actual MES/ERP test instance.
Test Data Input	Test instance connected; live export.

Expected Result	Data received and fields mapped correctly.	
Traceability	Relevant User Req.(s)	EIUR3
	Relevant System Req.(s)	EISR5
	Relevant Use Case.(s)	UC-002

Project Name	**AIBI CV for Manufacturing**	
Test Suit	System-Level Non-Functional	
Test Case ID	TC-EISR6-MAN-01	
What to Test	Field engineer configures workflow via config.	
Test Data Input	Field engineer, config files, and test job.	
Expected Result	System enforces new workflow correctly without code changes.	
Traceability	Relevant User Req.(s)	EIUR3
	Relevant System Req.(s)	EISR6
	Relevant Use Case.(s)	UC-005

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Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Functional	
Test Case ID	TC-SFR1-AUTO-01	
What to Test	QR detection returns all codes in a static image.	
Test Data Input	Multi-QR test image with N known codes.	
Expected Result	detect_qr_codes() returns N codes; payloads match ground truth.	
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	SFR1, SFR15, SFR16
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**

Test Suit	Automated Functional	
Test Case ID	TC-SFR2-AUTO-01	
What to Test	Field mapping assigns decoded values to logical fields.	
Test Data Input	Decoded QR list + workstation mapping config.	
Expected Result	Mapped JSON contains correct field-value assignments.	
Traceability	Relevant User Req.(s)	UFR2
	Relevant System Req.(s)	SFR2, SFR4
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Functional	
Test Case ID	TC-SFR4-AUTO-01	
What to Test	JSON builder aggregates decoded values + metadata.	
Test Data Input	Scan data object with decoded codes + metadata.	
Expected Result	One valid JSON object produced and schema-compliant.	
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	SFR4, SFR11, SFR12
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Functional	
Test Case ID	TC-SFR5-AUTO-01	
What to Test	Keyboard emulation builder outputs correct keystrokes.	
Test Data Input	Sample scan payload; delimiter config.	
Expected Result	Correct keystroke sequence returned.	
Traceability	Relevant User Req.(s)	UFR9
	Relevant System Req.(s)	SFR5

	Relevant Use Case.(s)	UC-001 , UC-005

Project Name	AIBI CV for Manufacturing	
Test Suit	Automated Functional	
Test Case ID	TC-SFR11-AUTO-01	
What to Test	JSON output matches published schema.	
Test Data Input	JSON outputs from multiple workflows.	
Expected Result	All outputs validate against schema.	
Traceability	Relevant User Req.(s)	UFR7
	Relevant System Req.(s)	SFR11
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing	
Test Suit	Automated Functional	
Test Case ID	TC-SFR12-AUTO-01	
What to Test	JSON output matches published schema.	
Test Data Input	JSON outputs from multiple workflows.	
Expected Result	All outputs validate against schema.	
Traceability	Relevant User Req.(s)	UFR4, UFR9
	Relevant System Req.(s)	SFR12, EISR3
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing
Test Suit	Automated Functional
Test Case ID	TC-SFR15-AUTO-01
What to Test	Decoder decodes all QRs in multi-code test image.
Test Data Input	Multi-code image with ground truth payloads.
Expected	All codes decoded; error rate ≤ spec.

Result		
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	SFR15
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Functional	
Test Case ID	TC-SFR16-AUTO-01	
What to Test	Output builder includes one entry per detected QR.	
Test Data Input	N detections from detection module.	
Expected Result	Output contains N entries, no duplicates.	
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	SFR16
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Functional	
Test Case ID	TC-SFR17-AUTO-01	
What to Test	Simulation injects synthetic QR codes into frames.	
Test Data Input	Recorded video and synthetic injection config.	
Expected Result	Detection pipeline sees synthetic codes at correct frames.	
Traceability	Relevant User Req.(s)	UFR11
	Relevant System Req.(s)	SFR17
	Relevant Use Case.(s)	UC-005

Project Name	**AIBI CV for Manufacturing**
Test Suit	Automated Functional
Test Case ID	TC-SFR18-AUTO-01

What to Test	Simulation events pass through keyboard-emulation pipeline.	
Test Data Input	Simulated scan events and mock buffer.	
Expected Result	Output keystroke sequence matches expected simulation values.	
Traceability	Relevant User Req.(s)	UFR11
	Relevant System Req.(s)	SFR18
	Relevant Use Case.(s)	UC-005

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Non-Functional	
Test Case ID	TC-PPSR2-AUTO-01	
What to Test	Real-time latency under system load.	
Test Data Input	Synthetic CPU/GPU load; scan input.	
Expected Result	Latency stays below threshold (e.g., <200 ms).	
Traceability	Relevant User Req.(s)	UFR1
	Relevant System Req.(s)	PPSR2
	Relevant Use Case.(s)	UC-001

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Non-Functional	
Test Case ID	TC-PPSR5-AUTO-01	
What to Test	QR detection accuracy at 720p resolution.	
Test Data Input	720p dataset with ground truth.	
Expected Result	Detection accuracy meets required threshold.	
Traceability	Relevant User Req.(s)	UFR3
	Relevant System Req.(s)	PPSR5
	Relevant Use Case.(s)	UC-001

Project Name	AIBI CV for Manufacturing	
Test Suit	Automated Non-Functional	
Test Case ID	TC-PDSR1-AUTO-01	
What to Test	System stores data locally (no external endpoints).	
Test Data Input	Config inspection script.	
Expected Result	All endpoints local; no external URLs.	
Traceability	Relevant User Req.(s)	UFR10, UFR7
	Relevant System Req.(s)	PDSR1
	Relevant Use Case.(s)	UC-002

Project Name	AIBI CV for Manufacturing	
Test Suit	Automated Non-Functional	
Test Case ID	TC-PDSR2-AUTO-01	
What to Test	No cloud dependencies in codebase.	
Test Data Input	Static code analysis.	
Expected Result	No cloud APIs referenced.	
Traceability	Relevant User Req.(s)	UFR10, UFR7
	Relevant System Req.(s)	PDSR2
	Relevant Use Case.(s)	UC-001, UC-002

Project Name	AIBI CV for Manufacturing	
Test Suit	Automated Non-Functional	
Test Case ID	TC-PDSR4-AUTO-01	
What to Test	Long-duration stability test.	
Test Data Input	8–24 hour continuous scan simulation.	
Expected Result	No crashes; stable memory/CPU profile.	
Traceability	Relevant User Req.(s)	UFR11

	Relevant System Req.(s)	PDSR4
	Relevant Use Case.(s)	UC-002

Project Name	AIBI CV for Manufacturing	
Test Suit	Automated Non-Functional	
Test Case ID	TC-ODSR1-AUTO-01	
What to Test	Only approved OSS/commercial libraries used.	
Test Data Input	Dependency audit script.	
Expected Result	All libraries approved; no unauthorized deps.	
Traceability	Relevant User Req.(s)	ODUR1
	Relevant System Req.(s)	ODSR1
	Relevant Use Case.(s)	UC-005

Project Name	AIBI CV for Manufacturing	
Test Suit	Automated Non-Functional	
Test Case ID	TC-OOSR4-AUTO-01	
What to Test	Legacy data imports successfully.	
Test Data Input	Legacy JSON/CSV files.	
Expected Result	Data loads without parsing errors.	
Traceability	Relevant User Req.(s)	OOUR3
	Relevant System Req.(s)	OOSR4
	Relevant Use Case.(s)	UC-004

Project Name	AIBI CV for Manufacturing
Test Suit	Automated Non-Functional
Test Case ID	TC-EISR3-AUTO-01
What to Test	Interoperability schema validation.
Test Data Input	JSON outputs from all workflows.

Expected Result	All validate against EISR schema.	
Traceability	Relevant User Req.(s)	EIUR2
	Relevant System Req.(s)	EISR3
	Relevant Use Case.(s)	UC-001, UC-002

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Non-Functional	
Test Case ID	TC-EISR5-AUTO-01	
What to Test	Automated export to MES/ERP.	
Test Data Input	Mock MES/ERP endpoint and sample payload.	
Expected Result	Payload accepted; fields correct.	
Traceability	Relevant User Req.(s)	EIUR3
	Relevant System Req.(s)	EISR5
	Relevant Use Case.(s)	UC-002

Project Name	**AIBI CV for Manufacturing**	
Test Suit	Automated Non-Functional	
Test Case ID	TC-EISR6-AUTO-01	
What to Test	Config-driven workflow control.	
Test Data Input	Two distinct config files; same scan sequence.	
Expected Result	Workflow behavior changes correctly based on config.	
Traceability	Relevant User Req.(s)	EIUR3
	Relevant System Req.(s)	EISR6
	Relevant Use Case.(s)	UC-005

Appendix TE

Project Name	AIBI CV for Manufacturing	
Test Case ID	TC-SFR1-AUTO-01	
Testing Tools Used	pytest, unittest.mock, numpy, cv2.QRCodeDetector	
Testing Type	Automated Unit Test	
Execution Steps:	1	Mock cv2.QRCodeDetector with 3 known QR codes
	2	Call AdvancedScanner.decode_qr() with test image (200x200)
	3	Verify return count equals 3 detections
	4	Compare decoded payloads: ["part_number:PN-001", "serial_number:SN-002", "batch_id:BATCH-003"]
	5	Assert all payloads match expected values exactly

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Nathan	2025-11-12	AdvancedScanner.decode_qr method not found	FAIL	decode_qr static method not implemented in AdvancedScanner	2025-11-24 by Nathan
2	Nathan	2025-11-24	Mock detector setup returning None values	FAIL	Mock configuration for cv2.QRCodeDetector returning None	2025-11-30 by Nathan
3	Nathan	2025-11-30	Single QR detection working correctly	PASS	-	-
4	Nathan	2025-12-01	test_single_qr_detection PASSED	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 50%, SFR1 tests passing after multiple iterations

Project Name	AIBI CV for Manufacturing
Test Case ID	TC-SFR2-AUTO-01

Testing Tools Used	pytest, ConfigManager, WorkstationConfig, BarcodeField	
Testing Type	Automated Unit Test	
Execution Steps:	1	Create WorkstationConfig with part_number and serial_number fields
	2	Process decoded QR list: ["part_number:PN-12345", "serial_number:SN-67890"]
	3	Execute AdvancedScanner.parse_barcode() for field mapping
	4	Verify mapped_data contains correct field-value pairs
	5	Assert part_number="PN-12345" and serial_number="SN-67890"
	6	Confirm no unmapped fields exist

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Jacob	2025-11-12	parse_barcode method not implemented	FAIL	parse_barcode static method missing from AdvancedScanner	2025-11-24 by Jacob
2	Nathan	2025-11-24	Field mapping logic working for colon format	PASS	-	-
3	Collin	2025-12-01	test_parse_colon_format PASSED	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, SFR2 field mapping tests all passing - colon and JSON formats working

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR4-AUTO-01	
Testing Tools Used	pytest, json module, datetime	
Testing Type	Automated Unit Test	
Execution Steps:	1	Create scan data with workstation_id, timestamp, and barcode data
	2	Build JSON object with workstation_id="workstation_01" and barcodes array
	3	Verify single JSON object with required schema fields
	4	Validate presence of workstation_id, timestamp, barcodes fields
	5	Assert JSON is serializable using json.dumps()

	6	Confirm metadata integrity and structure compliance

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Jacob	2025-11-12	JSON output missing required schema fields	FAIL	JSON schema definition incomplete with missing required fields	2025-11-24 by Jacob
2	Jacob	2025-11-24	Schema validation failing on data types	FAIL	Data type validation logic error in JSON builder	2025-11-30 by Jacob
3	Jacob	2025-11-30	JSON structure matches expected schema	PASS	-	-
4	Jacob	2025-12-01	test_build_complete_scan_json PASSED	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 50%, SFR4 JSON builder working after schema fixes

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR5-AUTO-01	
Testing Tools Used	pytest, unittest.mock	
Testing Type	Automated Unit Test	
Execution Steps:	1	Create scan payload: {"part_number": "PN-12345", "serial_number": "SN-67890"}
	2	Set field_order and delimiter="TAB" configuration
	3	Execute keystroke sequence builder logic
	4	Capture sequence: ["PN-12345", "TAB", "SN-67890", "TAB", "ENTER"]
	5	Compare against expected pattern with TAB delimiters
	6	Assert correct delimiter insertion between fields

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction

1	Collin	2025-11-12	Keyboard emulation logic not implemented	FAIL	Keystroke sequence generation logic not implemented	2025-11-24 by Collin
2	Daniel	2025-11-24	Keystroke sequence with TAB delimiters working	PASS	-	-
3	Jacob	2025-12-01	test_keystroke_sequence_generation PASSED	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, SFR5 keystroke sequence generation working with TAB delimiters					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR11-AUTO-01	
Testing Tools Used	pytest, unittest.mock, mock_open, json	
Testing Type	Automated Unit Test	
Execution Steps:	1	Create event: {"workstation_id": "workstation_01", "barcodes": [{"name": "part_number", "value": "PN-12345"}]}
	2	Mock file operations with save/load cycle
	3	Add audit metadata: saved_at, file_version during save
	4	Execute load operation and parse JSON data
	5	Compare core data integrity: workstation_id, barcodes match
	6	Verify audit metadata (saved_at, file_version) present

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Daniel	2025-11-12	File persistence methods not implemented	FAIL	save_event and load_event methods missing	2025-11-24 by Daniel
2	Daniel	2025-11-24	Mock file operations not working correctly	FAIL	mock_open configuration error in test setup	2025-11-30 by Daniel
3	Daniel	2025-11-30	Audit metadata not	FAIL	Audit metadata	2025-12-01 by Daniel

			being added to saved events		(saved_at, file_version) not implemented	
4	Daniel	2025-12-01	test_save_and_load_event PASSED	PASS	-	-
Execution Summary	Total executions: 4, Pass rate: 25%, SFR11 persistence working after multiple fixes					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR12-AUTO-01	
Testing Tools Used	pytest, json validation	
Testing Type	Automated Unit Test	
Execution Steps:	1	Generate outputs from manufacturing and quality workflows
	2	Validate required fields: workstation_id, timestamp, barcodes
	3	Check data types: workstation_id (str), timestamp (str), barcodes (list)
	4	Verify barcode structure: each has "name" and "value" fields
	5	Assert all workflow outputs pass schema validation
	6	Confirm no missing required fields in any output

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Jacob	2025-11-12	Schema validation logic not implemented	FAIL	JSON schema validation functions missing	2025-11-24 by Jacob
2	Jacob	2025-11-24	Data type validation failing for barcodes array	FAIL	Data type checking logic error for barcode objects	2025-11-30 by Jacob
3	Nathan	2025-11-30	Schema validation working for all workflows	PASS	-	-
4	Nathan	2025-12-01	test_scan_json_schema_compliance PASSED	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 50%, SFR12 schema validation working after fixes

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR15-AUTO-01	
Testing Tools Used	pytest, unittest.mock, numpy	
Testing Type	Automated Unit Test	
Execution Steps:	1	Mock 5 ground truth QR codes: ["code_1", "code_2", "code_3", "code_4", "code_5"]
	2	Execute AdvancedScanner.decode_qr() on 250x250 test image
	3	Count detected codes and verify all 5 codes found
	4	Compare detected set against ground truth set
	5	Assert error rate ≤ 1% specification threshold
	6	Confirm no missing required fields in any output

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Daniel	2025-11-12	Multi-code detection missing codes in mock	FAIL	Mock setup for detectAndDecodeMulti not configured properly	2025-11-24 by Daniel
2	Daniel	2025-11-24	All 5 ground truth codes detected correctly	PASS	-	-
3	Jacob	2025-12-01	test_no_missed_codes PASSED	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, SFR15 multi-code detection achieving 0% error rate

Project Name	**AIBI CV for Manufacturing**
Test Case ID	TC-SFR16-AUTO-01
Testing Tools Used	pytest, numpy, set-based deduplication
Testing Type	Automated Unit Test

Execution Steps:	1	Create raw detections with duplicates: 3 detections including 1 duplicate
	2	Process through deduplication logic using set operations
	3	Count unique entries in final output
	4	Verify count equals 2 (original 3 minus 1 duplicate)
	5	Check final output contains no duplicate QR codes
	6	Assert unique_detections contains only ["part_number:PN-001", "serial_number:SN-002"]

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Collin	2025-11-12	Deduplication logic not implemented	FAIL	Duplicate detection removal logic missing	2025-11-24 by Collin
2	Collin	2025-11-24	Set operations not removing duplicates correctly	FAIL	Set-based deduplicatio n algorithm error	2025-11-30 by Collin
3	Collin	2025-11-30	Deduplication still allowing some duplicates	FAIL	Remaining edge cases in deduplicatio n logic	2025-12-01 by Collin
4	Collin	2025-12-01	test_no_duplic ate_outputs PASSED	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 25%, SFR16 deduplication working after extensive debugging

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR17-AUTO-01	
Testing Tools Used	pytest, numpy, simulation framework	
Testing Type	Automated Unit Test	
Execution Steps:	1	Create 50-frame sequence using numpy arrays
	2	Define injection config: {10: "part_number:PN-SIM-001", 25: "serial_number:SN-SIM-002", 40: "batch_id:BATCH-SIM-003"}
	3	Execute simulation with synthetic QR injection at specified frames
	4	Process frames and record detection timing
	5	Verify codes detected at frames 10, 25, 40 exactly
	6	Assert detection timing matches injection configuration

Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Nathan	2025-11-12	Simulation injection framework not implemented	FAIL	Synthetic QR injection logic not implemented	2025-11-24 by Nathan
2	Nathan	2025-11-24	Frame timing not matching injection config	FAIL	Frame number to injection mapping error	2025-11-30 by Nathan
3	Collin	2025-11-30	Injection timing working correctly	PASS	-	-
4	Collin	2025-12-01	test_synthetic_qr_injection PASSED	PASS	-	-
Execution Summary	Total executions: 4, Pass rate: 50%, SFR17 injection working after timing corrections					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR18-AUTO-01	
Testing Tools Used	pytest, unittest.mock (keyboard library)	
Testing Type	Automated Unit Test	
Execution Steps:	1	Generate events: [{"field": "part_number", "value": "PN-SIM-123"}, {"field": "serial_number", "value": "SN-SIM-456"}]
	2	Mock keyboard.write and keyboard.press_and_release functions
	3	Process through keyboard-emulation pipeline
	4	Capture keystroke log: ["PN-SIM-123", "TAB", "SN-SIM-456", "TAB", "ENTER"]
	5	Compare against expected sequence with TAB delimiters
	6	Assert mock call counts: write=2, press_and_release=3

Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Jacob	2025-11-12	Simulation keyboard pipeline not implemented	FAIL	Simulation keystroke flow logic missing	2025-11-24 by Jacob

2	Jacob	2025-11-24	Mock call counts not matching expected values	FAIL	Mock keyboard call count validation error	2025-11-30 by Jacob
3	Jacob	2025-11-30	Keystroke sequence validation logic incorrect	FAIL	Sequence validation logic error in test	2025-12-01 by Jacob
4	Jacob	2025-12-01	test_simulation_keystroke_flow PASSED	PASS	-	-
Execution Summary	Total executions: 4, Pass rate: 25%, SFR18 keystroke pipeline working after extensive fixes					

Project Name	**AIBI CV for Manufacturing**					
Test Case ID	TC-PPSR2-AUTO-01					
Testing Tools Used	pytest, time.perf_counter, numpy					
Testing Type	Automated Unit Test					
Execution Steps:	1	Create 640x480 test image using numpy (white image)				
	2	Prepare scan input for AdvancedScanner.decode_qr()				
	3	Execute scan operation with timing measurement				
	4	Measure latency using time.perf_counter() in milliseconds				
	5	Verify latency < 200ms performance threshold				
	6	Assert latency meets real-time processing requirement				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Jacob	2025-11-12	Detection latency measured at 520ms	FAIL	QR detection algorithm performance too slow	2025-11-24 by Jacob
2	Jacob	2025-11-24	Latency improved to 310ms but still over threshold	FAIL	Processing pipeline still needs optimization	2025-11-30 by Jacob
3	Daniel	2025-11-30	Latency optimized to 45ms, under threshold	PASS	-	-

4	Daniel	2025-12-01	test_detection_latency_threshold PASSED	PASS	-	-
Execution Summary	Total executions: 4, Pass rate: 50%, PPSR2 performance meeting threshold after optimization					

Project Name	**AIBI CV for Manufacturing**					
Test Case ID	TC-PPSR5-AUTO-01					
Testing Tools Used	pytest, unittest.mock, numpy (720x1280 images)					
Testing Type	Automated Unit Test					
Execution Steps:	1	Create 720p dataset (720x1280) with 100 test cases				
	2	Mock 95% success rate through detectAndDecodeMulti				
	3	Execute AdvancedScanner.decode_qr() on all test images				
	4	Calculate accuracy: successes/total = 95/100 = 95%				
	5	Verify accuracy ≥95% threshold requirement				
	6	Assert error rate ≤1% meets specification				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Daniel	2025-11-12	Detection accuracy only 70% on 720p test cases	FAIL	Detection algorithm not optimized for 720p resolution	2025-11-24 by Daniel
2	Daniel	2025-11-24	Accuracy improved to 85%, still below 95% threshold	FAIL	Detection parameters need further tuning	2025-11-30 by Daniel
3	Daniel	2025-11-30	90% accuracy achieved, close but not meeting spec	FAIL	Final optimization needed to reach 95% target	2025-12-01 by Daniel
4	Daniel	2025-12-01	test_detection_accuracy_target PASSED	PASS	-	-
Execution Summary	Total executions: 4, Pass rate: 25%, PPSR5 accuracy meeting 95% threshold after multiple iterations					

Project Name	AIBI CV for Manufacturing	
Test Case ID	TC-PDSR1-AUTO-01	
Testing Tools Used	pytest, pattern matching, config inspection	
Testing Type	Automated Security/Compliance Test	
Execution Steps:	1	Load config data: {"database_url": "sqlite:///./data/scans.db", "output_directory": "./outputs"}
	2	Check for external patterns: ["http://", "https://", ".com", ".net", ".org"]
	3	Verify local patterns: ["./", "sqlite:///", "file://"] or numeric values
	4	Assert no external endpoints in any configuration values
	5	Confirm all storage paths are local-only
	6	Validate compliance with local data storage requirement

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Collin	2025-11-12	External URL pattern detected in config values	FAIL	External URL references found in configuration	2025-11-24 by Collin
2	Collin	2025-11-24	Still detecting .com patterns in config	FAIL	Pattern matching producing false positives	2025-11-30 by Collin
3	Collin	2025-11-30	Pattern matching rejecting valid local paths	FAIL	Pattern matching logic error for local paths	2025-12-01 by Collin
4	Collin	2025-12-01	test_local_endpoints_only PASSED	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 25%, PDSR1 local-only compliance achieved after config cleanup

Project Name	AIBI CV for Manufacturing	
Test Case ID	TC-PDSR2-AUTO-01	
Testing Tools Used	pytest, module inspection, pattern matching	
Testing Type	Automated Security/Compliance Test	
Execution Steps:	1	Import aibi_cv.advanced_scanner module for analysis

	2	Search for prohibited patterns: ["aws", "azure", "gcp", "boto3", "requests.post"]
	3	Inspect module dictionary and source for cloud API references
	4	Verify no cloud service patterns in module source
	5	Assert no prohibited cloud API patterns found
	6	Validate local-only processing compliance

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Daniel	2025-11-12	Found requests import in advanced_scanner module	FAIL	Requests and cloud API imports found in module	2025-11-24 by Daniel
2	Jacob	2025-11-24	No prohibited cloud imports detected	PASS	-	-
3	Nathan	2025-12-01	test_no_external_api_calls PASSED	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, PDSR2 no cloud API dependencies found in codebase

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-PDSR4-AUTO-01	
Testing Tools Used	pytest, simulation framework, stability monitoring	
Testing Type	Automated Stability/Endurance Test	
Execution Steps:	1	Initialize continuous scanning simulation loop
	2	Start extended scan simulation (8-24 hours)
	3	Monitor for exceptions, crashes, or memory leaks during operation
	4	Track system resource usage and error rates
	5	Analyze stability metrics for memory leaks or performance degradation
	6	Assert stable operation with no crashes throughout test duration

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Jacob	2025-11-12	System crashed after 2 hours of	FAIL	Memory management and	2025-11-24 by Jacob

			continuous operation		exception handling issues	
2	Jacob	2025-11-24	Memory leak detected after 6 hours	FAIL	Resource cleanup not implemented properly	2025-11-30 by Jacob
3	Daniel	2025-11-30	8-hour stability test completed successfully	PASS	-	-
4	Daniel	2025-12-01	24-hour endurance test passed	PASS	-	-
Execution Summary	Total executions: 4, Pass rate: 50%, PDSR4 stability achieved after memory leak fixes					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-ODSR1-AUTO-01	
Testing Tools Used	pytest, sys.modules inspection, dependency validation	
Testing Type	Automated Compliance Test	
Execution Steps:	1	Inspect sys.modules for loaded dependencies
	2	Compare against approved list: {"opencv-python", "numpy", "pytest", "keyboard", "pyzbar"}
	3	Check for prohibited commercial libraries with commercial indicators
	4	Verify no unauthorized dependencies in loaded modules
	5	Validate open-source license compliance
	6	Assert all libraries are approved for use

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Nathan	2025-11-11	Unauthorized commercial library found in dependencies	FAIL	Unauthorized commercial library in dependency list	2025-11-24 by Nathan
2	Nathan	2025-11-24	Commercial indicators still present in module names	FAIL	Commercial library references still present	2025-11-30 by Nathan

3	Nathan	2025-11-30	License validation logic not working correctly	FAIL	License validation algorithm error	2025-12-01 by Nathan
4	Nathan	2025-12-01	test_approved_dependencies_only PASSED	PASS	-	-
Execution Summary	Total executions: 4, Pass rate: 25%, ODSR1 dependencies approved after thorough cleanup					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-OOSR4-AUTO-01	
Testing Tools Used	pytest, json module, ConfigManager	
Testing Type	Automated Integration Test	
Execution Steps:	1	Create test JSON files with legacy WorkstationConfig format
	2	Execute ConfigManager.load_config() and JSON parsing functions
	3	Verify data loads without parsing errors or exceptions
	4	Validate parsed WorkstationConfig structure and BarcodeField objects
	5	Check data integrity: workstation_id, barcode_fields, camera_index
	6	Assert successful parsing of legacy configuration formats

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Nathan	2025-11-12	JSON parsing throwing exceptions on legacy format	FAIL	Legacy JSON format support not implemented	2025-11-24 by Nathan
2	Nathan	2025-11-24	Config structure validation rejecting valid configs	FAIL	Workstation Config validation logic error	2025-11-30 by Nathan
3	Collin	2025-11-30	Legacy parser working correctly	PASS	-	-
4	Collin	2025-12-01	All legacy JSON formats parsing successfully	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 50%, OOSR4 legacy import working after parser fixes

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-ETSR3-AUTO-01	
Testing Tools Used	pytest, datetime.fromisoformat, json validation	
Testing Type	Automated Integration Test	
Execution Steps:	1	Generate outputs from manufacturing and quality workflows
	2	Validate ETSR fields: workstation_id, timestamp, barcodes, schema_version
	3	Check timestamp format compliance using datetime.fromisoformat()
	4	Verify barcode structure: each dict has "name" and "value" fields
	5	Validate interoperability schema compliance across workflows
	6	Assert all workflow outputs pass ETSR validation requirements

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Collin	2025-11-12	ETSR schema validation logic not implemented	FAIL	ETSR schema validation logic missing	2025-11-24 by Collin
2	Collin	2025-11-24	Timestamp format not ISO 8601 compliant	FAIL	Timestamp format generation not ISO 8601 compliant	2025-11-30 by Collin
3	Collin	2025-11-30	Barcode structure missing required fields	FAIL	Barcode object structure validation missing fields	2025-12-01 by Collin
4	Collin	2025-12-01	test_interoperability_schema_validation PASSED	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 25%, ETSR3 interoperability compliance achieved after schema fixes

Project Name	**AIBI CV for Manufacturing**

Test Case ID	TC-ETSR5-AUTO-01	
Testing Tools Used	pytest, mock MES/ERP validation	
Testing Type	Automated Integration Test	
Execution Steps:	1	Create MES payload: {"source_system": "AIBI_CV_Scanner", "workstation_id": "workstation_01", "scan_data": {...}}
	2	Validate payload structure with required MES/ERP fields
	3	Check field correctness: source_system, workstation_id, timestamp, scan_data
	4	Verify data types and scan_data dictionary structure
	5	Simulate payload acceptance by mock MES/ERP system
	6	Assert export payload meets MES/ERP integration requirements

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Jacob	2025-11-12	MES payload missing required fields	FAIL	Required MES/ERP payload fields missing	2025-11-24 by Jacob
2	Jacob	2025-11-24	Field validation rejecting valid data types	FAIL	MES payload field validation logic error	2025-11-30 by Jacob
3	Daniel	2025-11-30	MES payload structure meets requirements	PASS	-	-
4	Daniel	2025-12-01	test_mes_payload_structure PASSED	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 50%, ETSR5 MES integration working after payload corrections

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-ETSR6-AUTO-01	
Testing Tools Used	pytest, ConfigManager, WorkstationConfig	
Testing Type	Automated Integration Test	
Execution Steps:	1	Create manufacturing config: part_number, serial_number required
	2	Create quality config: part_number, batch_id, inspector_id required

	3	Process identical scan sequence: ["part_number:PN-001", "serial_number:SN-002", "batch_id:BATCH-003"]
	4	Analyze required field differences and completion states
	5	Verify config1 != config2 required fields and different completion results
	6	Assert configs produce different, correct workflow behaviors

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Collin	2025-11-12	Config differences not affecting workflow behavior	FAIL	Config-driven workflow logic not implemented	2025-11-24 by Collin
2	Nathan	2025-11-24	Required field logic not working correctly	FAIL	Required field validation logic error	2025-11-30 by Nathan
3	Jacob	2025-11-30	Completion state validation logic incorrect	FAIL	Workflow completion state checking logic error	2025-12-01 by Jacob
4	Daniel	2025-12-01	test_different_config_workflows PASSED	PASS	-	-

Execution Summary	Total executions: 4, Pass rate: 25%, ETSR6 config-driven workflows working after logic fixes

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Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR1-MAN-01	
Testing Tools Used	Live camera, printed QR codes, system UI	
Testing Type	Manual Integration Test	
Execution Steps:	1	Set up live camera with 720p resolution
	2	Place printed QR codes in camera field of view
	3	Launch AIBI CV scanner application
	4	Observe real-time detection in UI
	5	Verify all visible QR codes are detected and displayed

	6	Count detected codes against physically visible codes

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Nathan	2025-11-15	Only 1 of 3 QR codes detected	FAIL	Camera focus calibration issue	2025-11-20 by Nathan
2	Nathan	2025-11-28	All 3 QR codes detected correctly	PASS	-	-
3	Jacob	2025-12-01	Consistent detection across multiple tests	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, SFR1 live camera detection working after hardware calibration

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR2-MAN-01	
Testing Tools Used	Configuration UI, printed test QR codes	
Testing Type	Manual Configuration Test	
Execution Steps:	1	Access workstation configuration interface
	2	Define field mapping: part_number, serial_number, batch_id
	3	Print test QR codes with known values
	4	Scan codes and verify field assignment in output
	5	Check that values appear in correct JSON fields
	6	Validate no field mismatches occur

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Collin	2025-11-15	Configuration UI missing	FAIL	UI component not implemented	2025-11-22 by Collin
2	Collin	2025-11-28	Fields mapped to correct JSON structure	PASS	-	-

3	Daniel	2025-12-01	All field assignments working correctly	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, SFR2 field mapping configuration functional					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR4-MAN-01	
Testing Tools Used	Live scanner, file system monitor	
Testing Type	Manual Integration Test	
Execution Steps:	1	Perform live scan with multiple QR codes
	2	Monitor outputs directory for JSON file creation
	3	Open generated JSON file and validate structure
	4	Verify presence of workstation_id, timestamp, barcodes array
	5	Check that all scanned data appears in correct format
	6	Confirm metadata integrity and completeness

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Jacob	2025-11-15	JSON structure invalid, missing required fields	FAIL	JSON schema validation error	2025-11-18 by Jacob
2	Jacob	2025-11-28	Valid JSON but timestamp field missing	FAIL	Timestamp generation bug	2025-11-29 by Jacob
3	Jacob	2025-12-01	Complete JSON with all required fields	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 33%, SFR4 JSON generation working after metadata fixes					

Project Name	**AIBI CV for Manufacturing**
Test Case ID	TC-SFR5-MAN-01
Testing Tools Used	Live scanner, Notepad application, keyboard monitoring

Testing Type	Manual Integration Test	
Execution Steps:	1	Open Notepad and position cursor in text field
	2	Configure scanner for keyboard emulation mode
	3	Perform live scan with test QR codes
	4	Observe keystroke output in Notepad
	5	Verify correct sequence: value1, TAB, value2, TAB, ENTER
	6	Check for accurate data transmission

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Daniel	2025-11-15	No keystroke output to target application	FAIL	Keyboard library not initialized	2025-11-19 by Daniel
2	Daniel	2025-11-28	Correct keystroke sequence generated	PASS	-	-
3	Nathan	2025-12-01	TAB delimiters and ENTER working properly	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, SFR5 keyboard emulation functional

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR11-MAN-01	
Testing Tools Used	Live scanner, network disconnect simulation	
Testing Type	Manual Persistence Test	
Execution Steps:	1	Disconnect network/downstream system
	2	Perform multiple scan operations while offline
	3	Verify scans are queued/stored locally
	4	Reconnect network/downstream system
	5	Observe automatic replay of queued events
	6	Confirm no data loss or duplicates

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction

1	Collin	2025-11-15	Scans lost when network disconnected	FAIL	No offline storage mechanism	2025-11-17 by Collin
2	Collin	2025-11-28	Queue created but data not replaying	FAIL	Replay trigger not working	2025-11-30 by Collin
3	Collin	2025-12-01	All offline scans replayed successfully	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 33%, SFR11 persistence working after queue implementation					

Project Name	**AIBI CV for Manufacturing**					
Test Case ID	TC-SFR12-MAN-01					
Testing Tools Used	Documentation review, JSON schema files, sample outputs					
Testing Type	Manual Documentation Test					
Execution Steps:	1	Review JSON schema documentation				
	2	Generate sample outputs from live system				
	3	Validate outputs against documented schema				
	4	Check for consistency between docs and implementation				
	5	Verify all required fields are documented				
	6	Confirm schema version matches implementation				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Nathan	2025-11-15	Schema docs don't match implementation	FAIL	Documentation version mismatch	2025-11-16 by Nathan
2	Jacob	2025-11-28	Documentation updated and consistent	PASS	-	-
3	Daniel	2025-12-01	Schema and output fully aligned	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, SFR12 documentation consistency achieved					

Project Name	AIBI CV for Manufacturing					
Test Case ID	TC-SFR15-MAN-01					
Testing Tools Used	Multi-code test sheet, live camera, typical factory lighting					
Testing Type	Manual Detection Test					
Execution Steps:	1	Print test sheet with 5 different QR codes				
	2	Set up typical factory lighting conditions				
	3	Position test sheet in camera field of view				
	4	Run live detection and count successful decodes				
	5	Verify all 5 codes are detected consistently				
	6	Test under various lighting angles				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Collin	2025-11-15	Only 3 of 5 QR codes detected under factory lighting	FAIL	Low light detection threshold too high	2025-11-21 by Collin
2	Collin	2025-11-28	All 5 codes detected consistently	PASS	-	-
3	Nathan	2025-12-01	Reliable detection across lighting conditions	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, SFR15 multi-code detection reliable under factory conditions					

Project Name	AIBI CV for Manufacturing	
Test Case ID	TC-SFR16-MAN-01	
Testing Tools Used	Live scanner, multiple QR codes, output verification	
Testing Type	Manual Output Verification Test	
Execution Steps:	1	Arrange multiple QR codes in camera view
	2	Perform live scan operation
	3	Count physical QR codes visible to camera
	4	Check output JSON for number detect codes
	5	Verify output count matches visible count

	6	Confirm no codes are missed or duplicated				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Jacob	2025-11-15	Output shows 2 codes when 3 visible	FAIL	Detection area boundary issue	2025-11-17 by Jacob
2	Jacob	2025-11-28	Output contains duplicate entries	FAIL	Duplicate filtering not implemented	2025-11-29 by Jacob
3	Jacob	2025-12-01	Output count matches visible codes exactly	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 33%, SFR16 output accuracy achieved after deduplication fixes					

Project Name	**AIBI CV for Manufacturing**					
Test Case ID	TC-SFR17-MAN-01					
Testing Tools Used	Simulation UI, recorded video footage, developer tools					
Testing Type	Manual Simulation Test					
Execution Steps:	1	Launch simulation mode interface				
	2	Load recorded video footage with known QR codes				
	3	Configure synthetic QR injection parameters				
	4	Run simulation and observe detection behavior				
	5	Verify simulation results match expected outcomes				
	6	Check simulation logs for accuracy				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Daniel	2025-11-15	Simulation interface not available	FAIL	DEF-012	Develop simulation UI components
2	Daniel	2025-11-28	Simulation executing with expected results	PASS	-	-

3	Collin	2025-12-01	Simulation behavior matches live system	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, SFR17 simulation functionality working					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-SFR18-MAN-01	
Testing Tools Used	Simulation mode, text editor, keyboard monitoring	
Testing Type	Manual Simulation Integration Test	
Execution Steps:	1	Enable simulation mode with keyboard emulation
	2	Open text editor for keystroke capture
	3	Run simulation with known test data
	4	Observe keystroke output in text editor
	5	Verify correct sequence and formatting
	6	Confirm simulation matches live behavior

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Nathan	2025-11-15	No keystroke output in simulation mode	FAIL	DEF-013	Integrate keyboard emulation with simulation
2	Nathan	2025-11-28	Keystrokes generated in wrong order	FAIL	DEF-014	Fix keystroke sequence logic
3	Nathan	2025-12-01	Correct keystroke sequence in simulation	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 33%, SFR18 simulation keyboard emulation functional					

Project Name	**AIBI CV for Manufacturing**
Test Case ID	TC-PUSR1-MAN-01

Testing Tools Used	New operator, system UI, stopwatch, task checklist	
Testing Type	Manual Usability Test	
Execution Steps:	1	Select operator with no prior system experience
	2	Provide basic system overview (5 minutes max)
	3	Start timer and begin basic workflow tasks
	4	Monitor operator progress without assistance
	5	Record completion time and error count
	6	Verify completion within 30-minute requirement

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Jacob	2025-11-16	New operator took 45 minutes to complete tasks	FAIL	DEF-015	Simplify UI workflow and add guided tutorial
2	Jacob	2025-11-29	Operator completed workflow in 25 minutes	PASS	-	-
3	Collin	2025-12-02	Different operator finished in 22 minutes	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, PUSR1 usability target met after UI improvements

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-PUSR2-MAN-01	
Testing Tools Used	System UI, test users, usability questionnaire	
Testing Type	Manual Usability Test	
Execution Steps:	1	Present system UI to test users
	2	Ask users to identify major controls and functions
	3	Evaluate tooltip effectiveness and clarity
	4	Test icon recognition without text labels
	5	Assess overall UI intuitiveness
	6	Record user feedback and success rates

Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Daniel	2025-11-16	Users misidentified 60% of status icons	FAIL	DEF-016	Redesign icons and improve color contrast
2	Daniel	2025-11-29	85% correct identification with tooltips	PASS	-	-
3	Nathan	2025-12-02	95% accurate state recognition	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, PUSR2 UI guidance working after tooltip improvements					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-PUSR3-MAN-01	
Testing Tools Used	Legacy system, new system, workflow comparison sheet	
Testing Type	Manual Workflow Comparison Test	
Execution Steps:	1	Document current legacy workflow steps
	2	Perform same task using new system
	3	Count manual input steps for each system
	4	Compare time required for task completion
	5	Evaluate reduction in repetitive actions
	6	Verify new system requires fewer steps

Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Collin	2025-11-16	New system: 8 steps vs Legacy: 12 steps	PASS	-	-
2	Jacob	2025-11-29	33% reduction in manual input steps	PASS	-	-
3	Daniel	2025-12-02	Workflow completion	PASS	-	-

			time reduced by 40%			
..						
Execution Summary	Total executions: 3, Pass rate: 100%, PUSR3 workflow improvement validated					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-PUSR4-MAN-01	
Testing Tools Used	Live system, operator observation, touch interaction counter	
Testing Type	Manual Touch-Free Operation Test	
Execution Steps:	1	Observe operator during typical work cycle
	2	Count number of UI touches/interactions required
	3	Monitor automatic detection and processing
	4	Verify minimal manual intervention needed
	5	Assess hands-free operation effectiveness
	6	Record touch frequency per cycle

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Nathan	2025-11-16	Operator required 8+ UI interactions per cycle	FAIL	DEF-017	Increase automation and reduce manual steps
2	Nathan	2025-11-29	Reduced to 3 interactions per cycle	PASS	-	-
3	Jacob	2025-12-02	Average 1.5 touches per cycle achieved	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, PUSR4 touch-free operation achieved					

Project Name	**AIBI CV for Manufacturing**
Test Case ID	TC-PUSR5-MAN-01
Testing Tools Used	Live system, consecutive job simulation, auto-fill monitoring
Testing Type	Manual Automation Test

Execution Steps:	1	Configure system for consecutive similar jobs
	2	Perform first job with full manual input
	3	Execute subsequent jobs and monitor auto-fill
	4	Verify common fields are automatically populated
	5	Check accuracy of auto-filled data
	6	Assess reduction in repetitive input

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Collin	2025-11-16	All fields require manual entry each time	FAIL	DEF-018	Implement auto-fill for common field values
2	Collin	2025-11-29	Auto-fill working for workstation and batch fields	PASS	-	-
3	Daniel	2025-12-02	70% reduction in repetitive data entry	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, PUSR5 automation reducing manual input effectively. Verify simulation results match expected outcomes

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-PUSR10-MAN-01	
Testing Tools Used	UI screenshots, test users, icon recognition test	
Testing Type	Manual Visual Design Test	
Execution Steps:	1	Prepare screenshots of system status indicators
	2	Present to users without text explanations
	3	Ask users to identify system states (OK/warning/error)
	4	Test color-coding effectiveness
	5	Evaluate icon clarity and recognition
	6	Record accuracy of state identification

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction

1	Jacob	2025-11-16	Users misidentifying error states as warnings	FAIL	DEF-019	Improve color coding and icon design
2	Jacob	2025-11-29	Color coding improved, 90% accuracy	PASS	-	-
3	Nathan	2025-12-02	95% accurate state identification achieved	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, PUSR10 visual indicators effective after color improvements					

Project Name	**AIBI CV for Manufacturing**					
Test Case ID	TC-PPSR2-MAN-01					
Testing Tools Used	Live system, operator, feedback, response time monitoring					
Testing Type	Manual Performance Test					
Execution Steps:	1	Have operator perform typical scanning tasks				
	2	Monitor system response times during operation				
	3	Ask operator about perceived delays				
	4	Test under normal factory load conditions				
	5	Verify "immediate" feedback perception				
	6	Record any noticeable delays				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Daniel	2025-11-16	Operator reported 2-3 second delays	FAIL	DEF-020	Optimize processing pipeline for real-time response
2	Daniel	2025-11-29	Response time under 500ms, feels immediate	PASS	-	-
3	Collin	2025-12-02	No perceived delays during operation	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, PPSR2 real-time performance achieved

Project Name	**AIBI CV for Manufacturing**					
Test Case ID	TC-PPSR5-MAN-01					
Testing Tools Used	720p camera hardware, live testing environment					
Testing Type	Manual Hardware Performance Test					
Execution Steps:	1	Set up system with actual 720p camera hardware				
	2	Test QR detection accuracy under factory conditions				
	3	Compare performance to development environment				
	4	Verify no degradation in detection rates				
	5	Test various lighting and distance conditions				
	6	Record accuracy metrics				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Nathan	2025-11-16	Detection accuracy dropped to 85% with 720p camera	FAIL	DEF-021	Calibrate detection parameters for 720p resolution
2	Nathan	2025-11-29	96% accuracy achieved with optimized settings	PASS	-	-
3	Jacob	2025-12-02	Consistent 720p performance across conditions	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, PPSR5 720p hardware performance validated					

Project Name	**AIBI CV for Manufacturing**
Test Case ID	TC-PDSR1-MAN-01
Testing Tools Used	Network monitoring tools, live system operation
Testing Type	Manual Security Test

Execution Steps:	1	Set up network traffic monitoring
	2	Run system through complete operational cycle
	3	Monitor all outbound network connections
	4	Verify no external data transmission occurs
	5	Check for any unauthorized network activity
	6	Confirm local-only operation

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Collin	2025-11-16	Found telemetry data being sent to external server	FAIL	DEF-022	Remove all external network calls and telemetry
2	Collin	2025-11-29	No outbound traffic detected during operation	PASS	-	-
3	Daniel	2025-12-02	Local-only operation confirmed over 8-hour test	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, PDSR1 local-only operation verified

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-PDSR2-MAN-01	
Testing Tools Used	Network disconnect, live system testing	
Testing Type	Manual Offline Operation Test	
Execution Steps:	1	Disconnect system from internet completely
	2	Attempt to run all normal operations
	3	Verify full functionality without network
	4	Test all scanning and processing features
	5	Confirm no degradation in performance
	6	Validate offline operation capability

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction

1	Jacob	2025-11-16	All functions work normally without internet	PASS	-	-
2	Nathan	2025-11-29	No network dependencies detected	PASS	-	-
3	Collin	2025-12-02	Offline operation maintained for 12 hours	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 100%, PDSR2 offline operation validated					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-PDSR4-MAN-01	
Testing Tools Used	Factory environment, extended operation monitoring	
Testing Type	Manual Stability Test	
Execution Steps:	1	Deploy system in actual factory environment
	2	Run continuous operation for 8+ hours
	3	Monitor for crashes, errors, or performance issues
	4	Test under typical factory conditions (dust, vibration, temperature)
	5	Record system stability metrics
	6	Verify acceptable performance throughout test

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Daniel	2025-11-17	System crashed after 4 hours due to memory leak	FAIL	DEF-023	Fix memory management and add monitoring
2	Daniel	2025-11-30	12-hour stable operation achieved	PASS	-	-
3	Nathan	2025-12-03	24-hour factory deployment successful	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, PDSR4 factory stability achieved

Project Name	**AIBI CV for Manufacturing**					
Test Case ID	TC-ODSR2-MAN-01					
Testing Tools Used	Documentation review, dependency analysis tools					
Testing Type	Manual Documentation Test					
Execution Steps:	1	Review all project documentation				
	2	Identify all open-source components used				
	3	Verify each component is properly documented				
	4	Check license information is complete				
	5	Confirm version numbers are accurate				
	6	Validate documentation completeness				
Test Execution Records:						
#	**Tester**	**Test Date**	**Actual Result**	**Status**	**Defect**	**Correction**
1	Jacob	2025-11-17	Missing license info for 3 OSS components	FAIL	DEF-024	Add complete license documentation
2	Jacob	2025-11-30	All OSS components documented with licenses	PASS	-	-
3	Collin	2025-12-03	Documentation complete and up-to-date	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, ODSR2 OSS documentation complete					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-OOSR4-MAN-01	
Testing Tools Used	Legacy system export files, import functionality	
Testing Type	Manual Data Migration Test	
Execution Steps:	1	Obtain actual legacy system export file
	2	Run data import process on real legacy data

	3	Verify all records import correctly
	4	Check data integrity after import
	5	Confirm no manual cleanup required
	6	Validate imported data accuracy

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Collin	2025-11-17	Import failed on 15% of legacy records	FAIL	DEF-025	Fix data format compatibility issues
2	Collin	2025-11-30	All 500 legacy records imported successfully	PASS	-	-
3	Daniel	2025-12-03	Data integrity validated, no corruption found	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, PUSR5 automation reducing manual input effectively

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-EISR4-MAN-01	
Testing Tools Used	Repository review, documentation audit, version control	
Testing Type	Manual Documentation Test	
Execution Steps:	1	Review repository for JSON schema files
	2	Check version control history for schema changes
	3	Verify documentation matches current schema version
	4	Confirm schema is properly tracked and versioned
	5	Validate documentation completeness
	6	Check for schema-documentation consistency

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Nathan	2025-11-17	Schema files not under version control	FAIL	DEF-026	Add schema files to git and implement versioning

2	Nathan	2025-11-30	Schema properly tracked with version tags	PASS	-	-
3	Jacob	2025-12-03	Documentation matches schema v1.2 exactly	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 67%, PUSR10 visual indicators effective after color improvements					

Project Name	**AIBI CV for Manufacturing**	
Test Case ID	TC-EISR5-MAN-01	
Testing Tools Used	MES/ERP test instance, live export functionality	
Testing Type	Manual Integration Test	
Execution Steps:	1	Set up connection to MES/ERP test instance
	2	Configure export parameters and field mapping
	3	Perform live scan and trigger export
	4	Verify data is received by MES/ERP system
	5	Check field mapping accuracy in received data
	6	Confirm successful integration

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Daniel	2025-11-17	Connection timeout to MES test server	FAIL	DEF-027	Configure network settings and retry logic
2	Daniel	2025-11-30	Connected but field mapping incorrect	FAIL	DEF-028	Fix field mapping configuration
3	Daniel	2025-12-03	MES integration working, all fields correct	PASS	-	-
..						
Execution Summary	Total executions: 3, Pass rate: 33%, ETSR5 MES integration functional after field mapping fixes					

Project Name	AIBI CV for Manufacturing	
Test Case ID	TC-EISR6-MAN-01	
Testing Tools Used	Configuration files, field engineer, test workflow	
Testing Type	Manual Configuration Test	
Execution Steps:	1	Have field engineer modify workflow configuration
	2	Change required fields and validation rules
	3	Deploy new configuration without code changes
	4	Test system with modified workflow
	5	Verify new workflow is enforced correctly
	6	Confirm configuration-driven behavior

Test Execution Records:

#	Tester	Test Date	Actual Result	Status	Defect	Correction
1	Jacob	2025-11-17	Config changes not reflected in system behavior	FAIL	DEF-029	Implement dynamic config reload mechanism
2	Jacob	2025-11-30	New workflow enforced after system restart	PASS	-	-
3	Collin	2025-12-03	Field engineer configured 3 workflows successfully	PASS	-	-
..						

Execution Summary	Total executions: 3, Pass rate: 67%, PPSR5 720p hardware performance validated